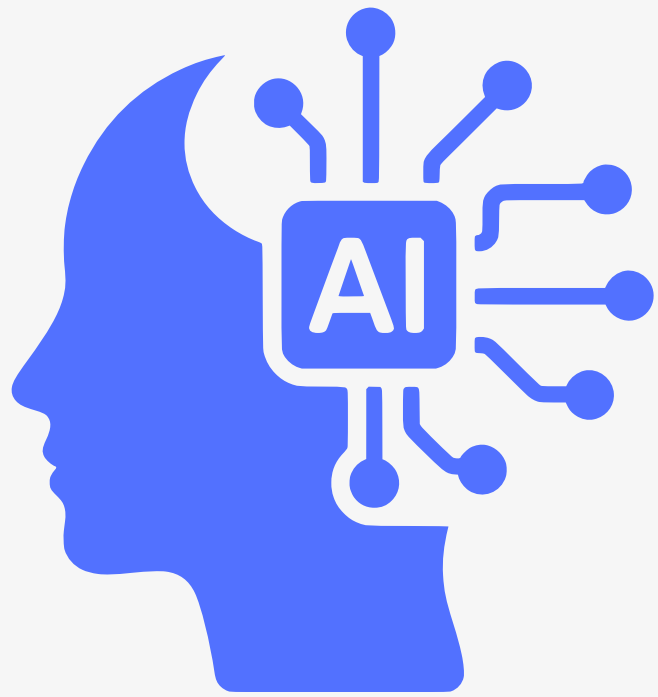




ASYNC PROCESSING SYSTEM

THE FASTEST AI APPLICATIONS DON'T WORK HARDER... THEY WORK
SMARTER BY PROCESSING TASKS IN THE BACKGROUND.





WHAT ARE QUEUES?

- Store tasks that need to be processed later.
- Execute tasks in the correct order.
- Prevent server overload.
- Improve application performance.
- Commonly powered by Redis.





WHAT ARE WORKERS?

- Background processes that execute queued tasks.
- Work independently from user requests.
- Handle heavy operations.
- Keep applications fast and responsive.
- Can process multiple jobs continuously.





WHAT ARE BACKGROUND JOBS?

- Long-running tasks executed after a request.
- Do not block the user interface.
- Improve user experience.
- Perfect for AI processing and automation.
- Run without user interaction.





REDIS QUEUES

- Redis stores jobs in memory.
- Extremely fast.
- Reliable job processing.
- Supports delayed and scheduled jobs.
- Widely used with BullMQ.





INGESTION WORKERS

- Automatically process uploaded data.
- Parse PDFs and documents.
- Extract text and metadata.
- Generate embeddings.
- Store processed data for AI retrieval.





RETRY SYSTEMS

- Retry failed jobs automatically.
- Handle temporary failures.
- Prevent data loss.
- Improve system reliability.
- Essential for production systems.





ASYNCHRONOUS PIPELINES

- Connect multiple background tasks.
- Each task triggers the next one.
- Fully automated workflows.
- Faster and scalable processing.
- Common in modern AI systems.

